Per Scholas

CAP 931 Capstone Report

Sales Account Assistant

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Overview 3

Technical Documentation 4

How the system works 4

Code structure 4

Key design decisions 5

Model selection 6

Guardrails 6

Inputs handling 7

Optional enhancements 7

Testing 8

Setup and running 9

Challenges and Solutions 11

Experiments 13

E1. Evidence gathering (search and parsing) 13

E2. One-pager design (four iterations) 13

E3. Prompt fixes found by reading live outputs 14

E4. Verifier outcomes (accuracy) 14

E5. Model split: cost and speed 15

Not yet run: single prompt vs. agent chain 15

System Outputs 17

Sample: Capital One 17

Sample: Target 25

Conclusion 32

Appendix 33

Production deployment plan.....33

Overview

I built a multi-agent prototype that helps a new sales rep prepare for a prospect account. The rep enters the product they sell, the prospect's website and their competitors. The app researches public sources and produces a one-page account brief: company strategy, competitor mentions, leadership, product and financial strategy, and talking points. Every claim links to its source.

Under the hood, the app collects public evidence about the prospect first, then runs a chain of specialized LLM agents over that evidence. A final verifier agent checks every statement against its cited source and removes anything the sources don't support.

Rubric criterion	Where it is addressed
Technical Setup	Technical Documentation: Setup and running
Inputs Handling	Technical Documentation: Inputs handling
LLM Model Selection & Use	Technical Documentation: Model selection, Guardrails · Experiments E2–E5
Data Integration & Output Relevance	Technical Documentation: How the system works · System Outputs
Optional Enhancements	Technical Documentation: Optional enhancements
Production Deployment	Appendix: Production deployment plan
Documentation Quality	This report: code structure, decisions, challenges, experiments, outputs

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Technical Documentation

How the system works

```
┌ Streamlit UI (app.py) ──────────────────────────────────────────────────────┐
│ product name · company URL · category · competitor URLs · value prop · target · deck │
└──────────────────────────────────────────────────────────────────────────────┘
                                        ▼
validation.py        URL normalization, required fields, length limits, ≤5 competitors,
                     competitor ≠ prospect, deck parsing (PDF/PPTX/DOCX/TXT)
                                        ▼
Intake agent         scope guardrail (refuses non-sales requests) + infers category and
                     search keywords from the product description and deck
                                        ▼
research/gather      1. prospect homepage + about/leadership/news/investor/careers pages
                     2. 9+ targeted searches: news, press, executive statements, leadership,
                        job postings, annual report, prospect×competitor pairs
                     3. filters out results that never mention the prospect
                     4. full-text extraction of top results (trafilatura)
                     5. SEC EDGAR: latest 10-K → Business, MD&A, keyword-relevant passages
                     → numbered evidence pool [1..n], each trimmed around prospect mentions
                                        ▼
Section agents       Strategy | Competitor Mentions | Leadership | Product & Financial
(parallel)           each sees only evidence types relevant to its job
                                        ▼
Writer agent         the one-page brief: chooses what matters for this sale and condenses
                     it; redrafts until it fits one printed page at a readable size
                                        ▼
Verifier agent       checks every claim against its cited evidence:
                     supported / partial (flagged ⚠) / unsupported (removed)
                                        ▼
render.py            one-page DOCX on the Per Scholas template (font sized to fit, citations
                     linked to sources) · brief + full research in app / Markdown · JSON
```

Code structure

```
src/cap_931/
  app.py             Streamlit UI (research form, results, downloads, alerts tab)
  validation.py      input validation and normalization
  pipeline.py        orchestration: intake → research → agents → writer → verifier
  agents.py          every agent prompt + guardrails
  llm.py             OpenAI wrapper: JSON output validated by Pydantic, retries, token accounting
  schemas.py         data models (inputs, evidence, sections, one-pager)
  render.py          Markdown (brief + full research) and the one-page DOCX
  layout.py          page-fit estimate shared by the writer agent and the DOCX renderer
  template.py        Per Scholas template helpers (cover, styles, Markdown → Word)
  assets/            perscholas_template.docx
  alerts.py          keyword alert watches (SQLite + SMTP)
  research/          fetch.py (scraping), search.py (Tavily/DuckDuckGo), edgar.py (10-K), deck.py,
gather.py
experiments/         prompt/model comparison harness
docs/                technical design and deployment plan
  samples/           a generated one-pager (Capital One) on the Per Scholas template
tests/               pytest suite (runs offline with a fake LLM)
```

docs/TECHNICAL.md covers the design decisions and model choice, and docs/DEPLOYMENT.md covers the production plan.

I kept the data flow strictly typed. Every step passes schemas.py models to the next one: `SalesInput → Intake → Evidence[] → section models → OnePageBrief → OnePager`.

Key design decisions

These are the main design choices I made, and why I made them.

Decision	Why
Gather the evidence first; the LLM only writes from it	LLMs make up executives, quotes and partnerships when asked about companies from memory. Every agent must cite evidence ids. The prompt treats a missing source as "not found", never as something to guess.
One agent per section, not one big prompt	Narrow prompts are easier to tune and test, and each agent gets a smaller, more relevant context. Sections run in parallel, so the chain adds little time. A direct comparison with a single prompt is set up in experiments/run_experiments.py (see Experiments).
The writer fits the brief to one page	The section agents gather everything; the writer decides what matters for *this* sale and how much space each section gets (no fixed counts). `layout.py` estimates the printed height (calibrated against Word); if the draft would overflow at 9pt, the writer is asked to cut it by the measured amount. The renderer then picks the largest font (10 → 9pt) that fits.
Verifier agent	Even grounded agents sometimes overstate. A second pass with the stronger model grades each claim against only its cited text. Unsupported claims are removed and partial ones are flagged. Citations to evidence ids that don't exist are dropped in code before the verifier runs.
Competitor mentions found in code first	A regex finds which evidence mentions each competitor before any LLM call. If none does, the section is filled without an LLM call, which removes the most common hallucination (made-up vendor relationships).
Search filter on the prospect's name	Search engines return loosely related results. Results whose title, snippet or text never mention the prospect are dropped.
SEC EDGAR for 10-Ks	Free, authoritative and structured. The parser skips the table of contents to find the real *Item 1. Business* and *Item 7. MD&A* sections, then scores paragraphs against the product's keywords.
Plain Python + Pydantic, no agent framework	Fewer dependencies and easier to explain and debug. Every agent output is a Pydantic model, and

	invalid JSON is retried with the validation error fed back to the model.
Disk cache (24h) for search, pages and filings	Repeat runs are faster and cheaper, and demos still work if a site starts rate-limiting. The alert job bypasses it.
DuckDuckGo fallback	The app works without a paid search key. Tavily is used when its key is set.

Model selection

Role	Default	Why
Intake, Strategy, Competitor, Leadership, Financial agents	`gpt-5-mini` (fast)	Extraction from supplied text is a well-bounded task. The small model handles it accurately at a fraction of the cost and latency, and these agents use most of the tokens (they read the evidence).
Writer, Verifier	`gpt-5` (smart)	Synthesis and fact-checking need stronger reasoning, and these calls use few tokens (section outputs, or only the cited excerpts), so the higher price adds little to the total cost.

- Reasoning-model calls use `reasoning_effort="low"` to keep latency down. Non-reasoning models get `temperature=0.2`.
- Both models can be changed in `.env` or in the sidebar. The sidebar lists every model the API key can access.
- Token usage per model is shown under **Run details** after each run, so cost can be checked.

Guardrails

- **Shared system prompt** (`agents.GUARDRAILS`). It:
  - limits every agent to account research;
  - requires citations;
  - forbids outside knowledge about the prospect;

- treats everything inside `<evidence>`, `<inputs>` and `<deck>` tags as untrusted data whose instructions are ignored. This defends against prompt injection from scraped pages.

- **Intake scope check:** requests unrelated to sales research (e.g. "write me a poem" in the value proposition) are refused before any research runs.
- **Input validation:** bad URLs, missing fields, oversized fields and too many competitors are rejected with a specific message per field.

Inputs handling

Field	Handling
Product name	required, ≤100 chars
Company URL	required. `https://` is added if missing, and the domain is validated. The company name comes from the site's schema.org data, then `og:site_name`, then the page title
Product category	optional, a word or a sentence. The intake agent normalizes it, or infers it from the description and deck
Competitor URLs	up to 5, separated by newlines or commas, de-duplicated, must differ from the prospect. Names are shortened to their brand ("Snowflake AI Data Cloud" → "Snowflake") for searching
Value proposition	required, 10–600 chars
Target customer	optional person or role. It's searched alongside the company, and findings appear under "About ⟨name⟩"
Deck upload	PDF, PPTX, DOCX, TXT or MD. Text is extracted and used by the intake agent

Optional enhancements

1. **Verifier agent (output improvement).** Grades every claim, removes unsupported ones, and flags partial ones.
2. **Keyword alert system.**
   - In the app's "Keyword alerts" tab, a rep sets up a watch: a company, keywords and an email address.

- `uv run cap-931-alerts` (cron or another scheduler) searches news and job postings for each watch.
- It stores already-seen URLs in SQLite and emails only new items over SMTP.
- "Check now" runs it on demand.

3. **Deck upload** to enrich the product context.

Using other model types (proposal)

- **Meeting deck generation:**
  - The one-pager JSON is already structured, so a deck generator can map each section to a slide using `python-pptx`.
  - An image model could generate the title visuals.
- **Rep-supplied documents** (RFPs, call notes, CRM exports):
  - Embed them with an embedding model.
  - Retrieve the relevant chunks into the evidence pool with their own source type, so they are cited like any other evidence.
- **Call preparation:**
  - A speech-to-text model transcribes previous calls.
  - The leadership agent can then add what each executive said in person.

Testing

`uv run pytest runs 31 tests offline`. A fake LLM returns schema-valid outputs, and the research layer is stubbed. They cover:

- the full pipeline, including verifier pruning and the out-of-scope refusal;
- competitor detection;
- the evidence budget;
- Markdown and DOCX rendering with hyperlinks;
- input validation;
- the HTML helpers and 10-K section parsing;
- deck extraction;

- alerts deduplication;
- LLM retry on invalid JSON;
- the Per Scholas template documents (cover filled, placeholders removed);
- the Streamlit UI end to end via `AppTest` (fill in the example → generate → downloads; validation errors; adding an alert).

Latest test run:

```
31 passed in 1.90s
```

Setup and running

The app requires Python 3.13 and uv.

```
uv sync                     # create .venv and install dependencies
cp .env.example .env        # then edit .env and set OPENAI_API_KEY
uv run cap-931              # opens the app at http://localhost:8501
```

Variable	Required	Purpose
`OPENAI_API_KEY`	yes	LLM access (can also be pasted in the app sidebar)
`OPENAI_FAST_MODEL` / `OPENAI_SMART_MODEL`	no	Defaults `gpt-5-mini` / `gpt-5`; the sidebar lists the models your key can use
`TAVILY_API_KEY`	no	Better search results; without it the app uses DuckDuckGo (no key)
`SEC_USER_AGENT`	no	Contact string the SEC asks EDGAR clients to send
`SMTP_USER` / `SMTP_PASSWORD`	no	Email delivery for keyword alerts (Gmail: use an app password)

1. Click **Fill in an example**, or enter your own product, prospect URL, competitor URLs, value proposition and target contact. Optionally upload a product deck (PDF, PPTX, DOCX, TXT).
2. Click **Generate account brief**. Progress is shown step by step (about 1–3 minutes).

3. Read the brief in the app, or download the **one-page brief** as .docx (Per Scholas template), or the brief plus full research as **.md**. The raw research is available as **.json**.

4. In **Keyword alerts**, watch a company for new news, press or job postings that match keywords.

Commands

```
uv run cap-931                                # web app
uv run cap-931-alerts                         # check all alert watches once (schedule with cron)
uv run pytest                                 # 31 tests: pipeline, agents, research, alerts,
documents, UI
uv run python experiments/run_experiments.py [url ...]   # prompt/model experiments (needs
OPENAI_API_KEY)
```

Challenges and Solutions

These are the main problems I ran into while building and testing the app, and how I solved each one.

Challenge	Solution
Search engines returned unrelated news (e.g. general market stories)	Results that don't mention the prospect are dropped.
DuckDuckGo returned "no results" when many queries ran in parallel	At most 2 concurrent queries, one retry, then a looser query without quotes or `OR`.
Competitor names scraped from sites were marketing names ("Snowflake AI Data Cloud") and matched nothing	The name is shortened to the brand word from the domain.
The 10-K parser first grabbed a cross-reference ("see Item 1. Business—Supervision…")	Take the first heading followed by more than 3,000 characters before the next item. This skips both the table of contents and later cross-references.
10-K text had no paragraph breaks, so relevance scoring failed	Insert line breaks at block elements before extracting text.
Long articles mention the prospect far below the fold	`focus_text` keeps paragraphs that mention the prospect or competitors first, within the character budget.
Models sometimes return JSON that doesn't match the schema	Pydantic validation, with the error sent back to the model, up to 3 attempts.
The "one-pager" came out 2–8 pages long	Make the writer agent choose the content, and add a layout estimate that measures the printed height. The writer redrafts until the brief fits (see Experiment E2).
Some lines switched font partway in Word	The model writes non-breaking hyphens (U+2011), which Word drew in a fallback font. Normalize those characters and set the font for every script slot.
The Per Scholas template numbered pages from 0, and its cover pushed content around	For the one-pager, start from the template's header and footer only, remove the page-numbering offset, and start content with "page break before" rather than relying on spacing.
A leader's "quote" was sometimes text *about* them	The leadership prompt now accepts only words the person said themselves.

The OpenAI credits ran out during testing and the app showed a raw error	Quota and rate-limit errors become plain messages with the billing link.

Experiments

Every experiment below was run on live data (real websites, Tavily/DuckDuckGo search, SEC EDGAR, OpenAI models). Each one records what I tried, what happened, and what I changed.

E1. Evidence gathering (search and parsing)

Experiment	Observation	Change	Outcome
Raw DuckDuckGo news for `"Capital One" Cloud Data Platform`	Most results were general market news that never mentioned the company	Drop results that don't mention the prospect's name	Irrelevant results removed
9 search queries in parallel	5 of 9 queries returned "no results" (rate limiting)	At most 2 concurrent queries, a retry, then a looser query	Sources found for Capital One: 17 → 48
Competitor query `"Capital One" "Snowflake AI Data Cloud"`	No results: the scraped marketing name matched nothing	Shorten names to the brand from the domain ("Snowflake")	Found the Capital One Software/Snowflake partnership and Databricks case studies
10-K section extraction	First version returned a cross-reference paragraph about liquidity	Take the first item heading followed by a long section	The real *Item 1. Business* overview and MD&A are extracted
DuckDuckGo vs Tavily	Both work. Tavily returns full page text and publication dates	Tavily when a key is set, DuckDuckGo as a no-key fallback	41–52 sources per account with Tavily

E2. One-pager design (four iterations)

The project brief asks for a one-pager. Getting a grounded brief to actually fit on one printed page took me four attempts. I rendered each version in Microsoft Word and counted the pages.

Version	Approach	Result
v1	Print all the research on the Per Scholas template (cover + every section)	8 pages: a report, not a one-pager
v2	Cover + an "at a glance" page made by cutting each list to its first 3–4 items	Overflowed by ⅓ page. Fixed cut-offs ignore what actually matters for the sale

v3	The writer agent writes the brief itself with a 380-word budget	275 words, still **2 pages**. Word counts ignore line wrapping, headings and citation links
v4	The writer agent redrafts until a **layout estimate** (calibrated against Word) says it fits at ≥9pt, and the renderer picks the largest font that fits	**1 page** for both Capital One (183 words) and Target (161 words), confirmed in Word

Takeaway: "fits on a page" is a layout constraint, not a word count. Measuring the layout and feeding the overflow back to the writer ("this draft is 18% too long") worked. Asking for N words did not.

E3. Prompt fixes found by reading live outputs

Problem seen in output	Cause	Prompt or code change	After
Category printed as "Cloud data platform / lakehouse (citations: [])"	The shared rule "cite every claim" made the intake agent, which has no evidence, write an empty citation	Intake prompt: "this step has no evidence; never write citations", plus a cleanup in code	Clean category ("Cloud data platform / Lakehouse", "Demand Forecasting")
Leaders' "quotes" such as "*Presented at AWS re:Invent 2024 by … Kajal Wood*"	The prompt asked for "a quote from the evidence", so the model quoted text *about* the person	"Only words this person said themselves… leave empty when the evidence only describes them"	2 of 7 leaders now have quotes, and both are genuine statements (Amy Lenander, Terren Peterson)
Dollar amounts garbled in the app ("$51.8 billion … $168.6 billion" shown as math)	Streamlit renders text between `$` signs as LaTeX	Escape `$` before display	Amounts display correctly
Talking points became generic under the tight v4 budget ("Offer cost control…")	Over-compression	Writer told to use the page well and tie each opener to a cited prospect action	**Not yet verified:** the OpenAI credits ran out before a re-run

E4. Verifier outcomes (accuracy)

The verifier grades every claim against the text of the source it cites.

Run	Claims checked	Supported	Partial (flagged ⚠)	Removed
Capital One, first live run	43	32	11	0
Capital One, sample in System Outputs	47	24	22	1
Target, sample in System Outputs	43	24	14	5

- **No claim reached the brief without a source.** Citations pointing to sources that don't exist are dropped in code, and claims whose sources don't support them are removed.
- **"Partial" usually means a condensed paraphrase.** The core fact is in the source but a detail (a year, a title) is not. Those claims stay in the brief with a ⚠ so the rep knows to double-check.

E5. Model split: cost and speed

The research agents run on the fast model (`gpt-5-mini`). The writer and verifier run on the smart model (`gpt-5`).

Run	Fast model: calls / tokens / time	Smart model: calls / tokens / time	Total time
Capital One	5 / 62.7k / 59s	6 / 65.9k / 85s	~2.5 min
Target	5 / 56.1k / 59s	6 / 53.7k / 86s	~2.5 min

- **The cheap model does the heavy reading.** The four section agents process about 55k prompt tokens of evidence, in parallel.
- **The expensive model gets only small inputs.** It sees the research summaries and the cited excerpts, not the whole evidence pool.
- **Running the section agents in parallel** keeps the four research steps to about one minute in total.

Not yet run: single prompt vs. agent chain

`experiments/run_experiments.py` compares three setups on the *same* evidence:

- a single-prompt baseline;
- the chain on the fast model only;
- the chain with the smart writer.

The same verifier scores all three. The harness is ready, but my OpenAI credits ran out during testing. Run it with `uv run python experiments/run_experiments.py` once credits are added.

System Outputs

Below are two real runs on different accounts and products. Each one shows the inputs, the generated one-pager, the full insights behind it (company strategy, competitor mentions, leadership, product and financial strategy), and every cited source, with links.

Sample: Capital One

FlowLake, a governed lakehouse (cloud data platform), sold to a large public bank.

Input	Value
Product name	FlowLake
Company URL	https://www.capitalone.com
Product category	Cloud data platform / lakehouse
Competitor URLs	https://www.snowflake.com, https://www.databricks.com
Value proposition	FlowLake unifies streaming and batch data in one governed lakehouse, cutting analytics infrastructure costs by up to 40% while meeting financial-services compliance requirements.
Target customer	Chief Data Officer

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Account Brief: Capital One

Selling FlowLake (Cloud data platform / Lakehouse) · Target: Chief Data Officer · Olanrewaju Muili · October 24, 2026

Capital One runs a governed lakehouse at scale on Snowflake and Databricks—target Amy Lenander with an interoperability and cost story.

Executive Summary

- CDO Amy Lenander owns enterprise data platforms and governance decisions. [1] [2] ⚠
- Snowflake has been foundational since 2017 for large-scale analytics and ML. [3] [4] ⚠
- They use lakehouse patterns with Apache Iceberg and Delta Lake for interoperability. [5] [6]

Company Strategy

- Cloud-first lakehouse supports near-real-time analytics, AI/ML, and regulatory reporting. [7] [5] ⚠
- Open table formats provide transactional guarantees and write-once, read-anywhere access. [5] [6]

Competitor Landscape

- Snowflake is entrenched as a core platform with public case studies. [3] [4] ⚠
- Capital One Software deepened Snowflake collaboration, including Databolt tokenization and Slingshot. [8] [9]
- Databricks powers a centralized Feature Hub using Photon and Delta Lake. [10]

Key Contacts

- **Amy Lenander**, Chief Data Officer: Primary owner of data platforms and governance. [1] [2] ⚠
- **Richard Scott Blackley**, Chief Financial Officer: Evaluates platform TCO and budget approvals. [11] ⚠
- **Frank G. LaPrade, III**, Chief Enterprise Services Officer: Stakeholder for shared platforms and governance. [11] ⚠

Financial Snapshot

- Public company; 2025 10-K notes Discover acquisition with $51.8B consideration. [12]

Talking Points

- Unify streaming and batch in one governed lakehouse for regulatory reporting. [7] [5] ⚠
- Complement existing Snowflake and Databricks with Iceberg/Delta interoperability. [5] [6] ⚠
- Offer cost control aligned to their warehouse versus lakehouse tradeoff focus. [7] [6] ⚠

Sources: [1] capitalone.com [2] cdomagazine.tech [3] snowflake.com [4] snowflake.com [5] capitalonesoftware.com [6] capitalone.com [7] capitalone.com [8] capitalonesoftware.com [9] capitalonesoftware.com [10] databricks.com [11] comparably.com [12] sec.gov · *Each claim was checked against its source by a verifier agent (⚠ = partly supported). Full research is in the app and its Markdown/JSON downloads.*

1

Figure: the generated one-pager for Capital One (every [n] links to its source).

Run statistics: 41 sources collected, 27 cited · 46 claims kept after verification (24 fully supported, 22 partly supported) · 1 claim removed by the verifier · 128,597 tokens · about 3 minutes of model time.

Insights for Capital One: full research behind the one-pager

Company Strategy

Capital One has prioritized a cloud-first, lakehouse-oriented data strategy focused on unified analytics, governance, and AI/ML at scale. The company (led by CDO Amy Lenander) has adopted Snowflake as a core platform while implementing open table formats (Apache Iceberg and Delta Lake) and lakehouse patterns to enable interoperable, governed data for analysts, engineers and ML workloads.

Initiatives & public statements

- Amy Lenander is Capital One's Chief Data Officer and leads the enterprise data strategy, building and managing data platforms and data management practices across the company. [4][5][6]
- Capital One completed a multi-year cloud migration (closing on-premises data centers) and adopted Snowflake beginning in 2017; Snowflake is described as a foundational step in Capital One's cloud data journey and is used to scale data operations and ML workloads (e.g., training models for Eno). [20][27] ⚠
- Capital One has implemented lakehouse architectures using open table formats (Apache Iceberg and Delta Lake) to provide transactional guarantees, schema enforcement, versioning, and cross-engine interoperability for governed analytics. [18][21][17]
- Capital One Software (the company's B2B software arm) is deepening collaboration with Snowflake, including integrations like Databolt tokenization inside Snowflake and a Slingshot feature to help customers choose compute options; Capital One states it adopted Snowflake in 2017 and leverages Snowflake for governance and scale. [26][20] ⚠
- Capital One's public technical content frames lakehouse designs as a way to support near-real-time analytics, AI/ML, and regulatory reporting while emphasizing cost and governance tradeoffs between warehouses and lake/lakehouse approaches. [17][21] ⚠

Technology & hiring signals

- Capital One publishes engineering content and presentations describing use of Apache Iceberg (and Delta Lake) to implement lakehouse architectures and to enable 'write once, read anywhere' interoperability across compute engines. [18][21]

- Capital One Software has built integrations that run inside Snowflake (Databolt tokenization using Snowpark Container Services and the Snowflake Native App Framework) indicating in-Snowflake compute and native app usage patterns. [26]
- Capital One positions Snowflake as a primary production platform for data science and analytics workloads (self-service Snowflake resources, separation of storage/compute, and ML model training). [27][20] ⚠
- Capital One's published materials explicitly call out supporting near-real-time analytics, AI/ML and regulatory reporting as drivers for lakehouse architecture choices. [17][18]

Competitor Mentions

Capital One has active, multi-faceted relationships with both Snowflake and Databricks: Snowflake is a long-standing platform/customer and partner (including Capital One Software product integrations), while Databricks is a customer and strategic engineering platform used for high-performance feature and pipeline work; both vendors' technologies and skills appear in Capital One job postings and public case studies.

Snowflake: ✅ mentioned

- Capital One is a customer of Snowflake and appears in Snowflake-hosted case material describing how Capital One uses Snowflake for cloud-scale data, ML model training, governance and to support products such as Eno. [20][27]
- Capital One Software (the enterprise B2B arm of Capital One) has a deepened collaboration/partnership with Snowflake, announced June 3, 2025, including integrations such as Databolt tokenization for Snowflake and Slingshot optimization tooling. [26][30]
- Capital One executives presented at/participated in Snowflake events and joint media (Snowflake Data Cloud Summit presentation and a video interview about the Snowflake partnership). [18][28]
- Snowflake and Capital One Ventures co-invested in a third party (Hightouch), indicating a financial/strategic interaction between Snowflake and Capital One Ventures. [29]
- Capital One job postings list Snowflake as a required or evaluated platform/skill for data engineering and related roles. [13][14][15][16]

Databricks: ✅ mentioned

- Capital One is a customer of Databricks and has a published Databricks customer story describing a centralized Feature Hub built on Databricks (Photon compute and Delta Lake), with reported performance and cost gains. [34]

- Capital One (alongside Nvidia) invested in Databricks in a funding round (Capital One backed Databricks at a reported $43 billion valuation). [35]
- Capital One job postings list Databricks (and Spark) as required or evaluated platforms/skills for multiple data engineering roles. [13][14][15][16]
- Capital One's public technical content traces the industry evolution that highlights Apache Spark and Databricks as key technologies for unified analytics and lakehouse architectures. [17][19][21]

Leadership

- **Richard D. Fairbank**, Founder; Chairman and Chief Executive Officer: As founder, Chairman and CEO he oversees Capital One's strategic direction and management of day-to-day operations and would sponsor or sign off on large strategic IT/data investments and vendor relationships. [9][8][10] ⚠
- **Amy Lenander**, Chief Data Officer: Leads Capital One's enterprise data strategy and the team that builds and manages data platforms and products — primary budget and decision owner for data platforms, governance, and analytics infrastructure. [4][5][6][7][24][23] ⚠ > "building a great data ecosystem is a huge enabler to powering transformative experiences for customers."
- **Richard Scott Blackley**, Chief Financial Officer: As CFO he controls or influences capital and operating budgets; would be involved in evaluating cost savings claims such as infrastructure reductions and total cost of ownership. [8] ⚠
- **Frank G. LaPrade, III, J.D.**, Chief Enterprise Services Officer: Responsible for enterprise services (likely including shared platforms and operations) and would be a stakeholder for infrastructure, platform consolidation, and enterprise governance. [8] ⚠
- **Sanjiv Yajnik**, President of Financial Services: Leads the financial-services business whose teams consume analytics and must meet compliance and auditability requirements; a stakeholder for data-driven product features and regulatory controls. [8] ⚠
- **John G. Finneran, Jr.**, General Counsel and Corporate Secretary: Head of legal and compliance matters; important for review of governance, data protection, regulatory compliance and contractual risk for data platforms used in financial services. [8] ⚠

About Chief Data Officer

- The Chief Data Officer at Capital One (Amy Lenander) spearheads the development and delivery of the company's data strategy, leads the central enterprise data team that builds and manages data platforms and products, and focuses on building a trustworthy, governed data ecosystem to power AI and business use cases. [4][6][7][23][24]

Publicly traded (SEC filer).

- Capital One files SEC reports and published a Form 10-K for fiscal year ended December 31, 2025, describing itself as a diversified financial services holding company and noting material corporate activity (the May 18, 2025 acquisition of Discover); the 10-K lists the fair value of purchase consideration for the Discover acquisition as $51.8 billion, total identifiable assets acquired as $168.6 billion (including $108.2 billion of loans held for investment), and deposits assumed of $106.9 billion. [41]
- Capital One states it was the largest issuer of credit cards in the U.S. by outstanding balance of credit card loans as of December 31, 2025 and serves customers through digital channels and a large mix of distribution channels (branches, cafés, call centers, ATMs), indicating large scale and significant payments transaction volume. [41] ⚠
- Capital One has an explicit enterprise data strategy and an enterprise data leadership team: Amy Lenander is Chief Data Officer (CDO), leading the central enterprise data team to build and manage data platforms and products across the company; her role began in February 2023 and she emphasizes making data easy to use and govern across business stakeholders. [4][5][6][7] ⚠
- Capital One has a history of adopting lakehouse architectures and open table formats: the company published guidance on evolving data lakes, described implementing a lakehouse on Snowflake using Apache Iceberg/Delta Lake, and emphasizes open table formats (Delta Lake, Apache Iceberg) and ‘write once, read anywhere’ interoperability as part of its platform strategy. [17][18][21][19]
- Capital One is an active, large-scale customer of both Snowflake and Databricks: public materials note Capital One uses Snowflake as a core Data Cloud platform (citing nearly 11,000 tech employees and a completed cloud migration) and has an operational partnership with Databricks (including a managed Disaster Recovery offering and failover capability for lakehouse analytics). [20][36]
- Capital One regularly explores acquisitions of technology companies and related assets to improve IT infrastructure and deliver on its digital strategy, and may fund acquisitions with equity or debt, signaling an ongoing appetite to invest in data/platform capabilities. [41]

Sources for Capital One

#	Source	Type	Published
4	Interview with Amy Lenander, Chief Data Officer | Capital One	News	-
5	Amy Lenander - Capital One | LinkedIn	News	-

6	Inside Capital One's Data Transformation: A Conversation with CDO Amy Lenander - CDO Magazine	News	-
7	Amy Lenander, Chief Data Officer, Capital One - DataIQ	News	-
8	Capital One Executive Team | Comparably	Leadership	-
9	Our Company | Capital One	Leadership	-
10	Our Board of Directors | Capital One Financial Corp.	Leadership	-
13	Data Engineer 4 (Manager, IC) - Capital One | Built In	Jobs	-
14	Best Data Engineer Jobs in NYC, NY 2026 | Built In NYC	Jobs	-
15	Senior Director, Data Engineer at Capital One	Jobs	-
16	Data Engineer 4, Real-time Data Platform at Capital One	Jobs	-
17	Understanding the Evolution of Data Lakes | Capital One	Web	-
18	Using Iceberg Tables to Implement a Lakehouse Architecture | Capital One Software	Web	-
19	Understanding the evolution of data lakes	Web	-
20	How Capital One Leverages the Power of Data in the Cloud	Web	-
21	Lakehouse Convergence: Delta Lake & Iceberg | Capital One Tech	Web	-
23	Accelerating AI with Data Management | Capital One	Leadership	-
24	Targeting AI: How Capital One is building an AI-ready data ecosystem with creative talent	Leadership	-
26	Capital One Software Deepens Collaboration with Snowflake | Capital One Software	Web	-
27	Capital One Banks on Snowflake for Data Science	Web	-
28	Inside The Capital One Software And Snowflake Partnership	Web	-
29	Snowflake and Capital One Ventures invest in Hightouch's AI platform	Web	-
30	Snowflake + Capital One Software	Web	-
34	Transforming data pipelines to power faster credit decisions | Databricks	Web	-
35	Nvidia and Capital One invest in Databricks at $43 billion valuation	Web	-

36	How Databricks Managed Disaster Recovery Helps Capital One Achieve Lakehouse Resilience | Databricks Blog	Web	-
41	CAPITAL ONE FINANCIAL CORP Form 10-K (filed 2026-02-19)	SEC filing	2026-02-19

Sample: Target

ShelfSense, AI demand forecasting, sold to a national retailer (category left blank, inferred).

Input	Value
Product name	ShelfSense
Company URL	https://www.target.com
Product category	(blank: inferred as Demand Forecasting)
Competitor URLs	https://blueyonder.com, https://o9solutions.com
Value proposition	ShelfSense uses AI demand forecasting to cut out-of-stocks and excess inventory, improving store-level availability while lowering carrying costs.
Target customer	Chief Supply Chain Officer

Per Scholas

Account Brief: Target

Selling ShelfSense (Demand Forecasting) · Target: Chief Supply Chain Officer · Olanrewaju Muili · October 24, 2026

Target is elevating AI-driven demand forecasting under new supply-chain leadership to fix in-stock issues and improve store reliability.

Executive Summary

- Target is using AI to improve demand forecasting to reduce in-stock issues. [1]
- Jeff England was hired as Chief Supply Chain Officer to accelerate reliability. [2] [3]

Company Strategy

- New food DC and Houston receive center expand capacity for hard-to-forecast goods. [1]
- Target is hiring a Director of Data Sciences for supply-chain planning. [4]
- CIO publicly emphasizes Generative AI to transform retail experiences. [5]

Competitor Landscape

- o9 hired ex-Target executive Ashwin Rao, creating a personnel link. [6] [7]

Key Contacts

- **Jeff England**, EVP & Chief Global Supply Chain and Logistics Officer: Primary buyer for forecasting and inventory reliability. [2] [3] ⚠
- **Lisa Roath**, Chief Operating Officer: Operations leader championing AI forecasting improvements. [1]
- **Michael J. Fiddelke**, Chief Executive Officer: Prioritizes merchandising, guest experience, and technology acceleration. [8]

Financial Snapshot

- Net sales were $104.8B in 2025; stores fulfilled over 97% of sales. [9] [8]
- Target plans about $5B investment, including ~$1B incremental in improvements. [10]

Talking Points

- You're using AI to cut volatility causing in-stock issues—let's target store-level accuracy gains. [1] ⚠

Sources: [1] finance.yahoo.com [2] prnewswire.com [3] corporate.target.com [4] corporate.target.com [5] youtube.com [6] dmagazine.com [7] bwdisrupt.com [8] corporate.target.com [9] sec.gov [10] wsj.com · *Each claim was checked against its source by a verifier agent (⚠ = partly supported). Full research is in the app and its Markdown/JSON downloads.*

1

Figure: the generated one-pager for Target (every [n] links to its source).

Run statistics: 52 sources collected, 22 cited · 38 claims kept after verification (24 fully supported, 14 partly supported) · 5 claims removed by the verifier · 109,823 tokens · about 3 minutes of model time.

Insights for Target: full research behind the one-pager

Company Strategy

Target is prioritizing supply-chain and store-level reliability as part of a turnaround under new CEO Michael Fiddelke, investing in stores, staffing and technology. Executives cite use of AI to improve demand forecasting and Target is making supply-chain network investments (new DCs/receive centers) while hiring senior supply-chain leadership to accelerate execution.

Initiatives & public statements

- Target executive Lisa Roath said the company is working to use AI to improve demand forecasting to reduce volatility that leads to in-stock issues. [2]
- Target announced the hiring of Jeff England as Chief Supply Chain Officer to accelerate supply-chain improvements and inventory reliability. [2] ⚠
- New CEO Michael Fiddelke has outlined priorities including leading with merchandising authority, elevating the guest experience, accelerating technology, and strengthening teams; company communications under Fiddelke emphasize improving the customer experience and store operations. [21][8]
- Target is shifting investment toward stores and store staffing (adding front-line labor and guest-experience training) while eliminating about 500 roles at distribution centers and regional offices to fund those investments. [8][9][10]
- Target has recently opened new supply-chain facilities—a food distribution center in Colorado and a receive center in Houston—to expand upstream capacity and buffer regional distribution for seasonal or hard-to-forecast goods. [2]

Technology & hiring signals

- Target's Executive Vice President and CIO (Brett Craig) has publicly discussed Generative AI and the role of Target's technology organization in leveraging AI to transform retail and improve guest and employee experiences. [18][19]
- Target is hiring for demand-planning and forecasting roles (demand planner listings) and has an open Director of Data Sciences — Supply Chain Network Planning and Design role focused on analytical models and decision-support systems. [27][30]

Competitor Mentions

o9 Solutions is connected to Target in the public record through a former Target executive (Ashwin Rao) who joined o9; there is no evidence in the provided documents that Blue Yonder has a direct relationship with Target.

Blue Yonder: — no mention found

o9 Solutions: ✅ mentioned

- o9 Solutions has hired Ashwin Rao into senior technical leadership roles; Rao previously held executive positions at Target, creating a personnel/link connection between o9 and Target. [46][51]

Leadership

- **Michael J. Fiddelke**, Chief Executive Officer, Target Corporation: CEO driving Target's turnaround and prioritizing merchandising, guest experience and supply-chain improvements that affect demand forecasting and on-shelf availability. [22][24] ⚠ > "It's the start of a new chapter for Target and we're moving quickly to take action against our priorities that will drive growth within our business."

- **Jeff England**, Executive Vice President and Chief Global Supply Chain and Logistics Officer (Chief Supply Chain Officer): Leads Target's end-to-end supply chain, responsible for replenishment, fulfillment, distribution and accelerating supply-chain plans — primary buyer/stakeholder for demand-forecasting solutions that improve store-level availability and reduce out-of-stocks. [36][37][38] ⚠ > "It's an honor to join Target and be part of such an iconic brand. I'm confident in the company's growth plan and priorities, and I'm excited to get started with the team, building on its strong foundation to further strengthen the speed, reliability and precision of the supply chain."

- **Lisa Roath**, Chief Operating Officer, Target Corporation: COO overseeing end-to-end retail operations including supply chain and stores; she links supply-chain improvements to sales and inventory reliability — relevant for operational adoption and ROI of demand-forecasting tools. [22][2] ⚠ > "We're working to use AI to improve our demand forecasting, which helps reduce some of the volatility that can lead to some of those in-stock issues."

- **Cara Sylvester**, Chief Merchandising Officer, Target Corporation: Top merchandising executive responsible for assortment design, product development and merchandising authority — stakeholder in store-level demand forecasts and inventory allocation decisions. [22] ⚠

- **Gretchen McCarthy**, Strategic Advisor (former Executive Vice President & Chief Supply Chain and Logistics Officer): Former chief supply chain officer who led inventory and operations; recently transitioned to strategic advisor and authored internal communications about store investments and supply-chain changes — institutional knowledge and potential internal

sponsor/ally on forecasting initiatives. [36][39][8] ⚠ > "As my Target chapter winds down this week, I'm filled with so much gratitude for the past 22 years. When I first joined as a buyer in Baby,..."

- **Adrienne Costanzo**, Chief Stores Officer, Target Corporation: Leads store operations and staffing investments; involved in decisions to increase front-line labor and guest-experience training — relevant to store-level execution of inventory and on-shelf availability improvements driven by forecasting. [8][10] ⚠ > "This change also fuels our ability to put significantly more payroll in our stores—primarily in additional labor and hours where needed most, but also in new guest experience training for every team member at every store,"
- **Brian C. Cornell**, Executive Chair, Target Corporation: Executive chair of the board; provides governance and oversight during the company's turnaround and investment decisions that fund supply-chain and technology programs. [24][25] ⚠
- **Katie M. Boylan**, Executive Vice President & Chief Communications Officer: Leads communications for Target; relevant for coordinating external/internal messaging around supply-chain initiatives and vendor partnerships. [25] ⚠

About Chief Supply Chain Officer

- Target's chief supply chain role (Chief Global Supply Chain and Logistics Officer / Chief Supply Chain Officer) is held by Jeff England, who joined in 2026 and reports to COO Lisa Roath. He is responsible for accelerating Target's supply chain plans and focused on improving reliability, speed and precision through technology, engineering and operational excellence. [36][37][38]
- Former chief supply chain officer Gretchen McCarthy transitioned to a strategic advisor role through August 2026 and co-authored internal communications about investing in stores and supply-chain changes. [36][8][39] ⚠

Product & Financial Strategy

Publicly traded (SEC filer).

- Target is a single operating segment retailer with scale: more than ~2,000 stores, more than 60 supply-chain facilities, and a workforce of more than 400,000 team members; merchandise sells across stores and digital channels and stores fulfilled more than 97% of total merchandise sales in each of the last three years. Reported Net Sales were $104.8B (2025), $106.6B (2024) and $107.4B (2023, 53 weeks). [52][21]
- Target's stated corporate strategy centers on four priorities that include 'Lead with Merchandising Authority', 'Elevate the Guest Experience', and 'Accelerate Technology to Enable Our Team and Delight Our Guests' — explicitly calling out investments in technology, data and operational capabilities to enable personalization and improve execution. [52][21]

- Target is actively investing in supply chain capacity, facilities and data/analytics to improve inventory reliability: recent actions include opening a new food distribution center and a first-ever receive center (Houston) expected to process ~25 million cartons/year, plus stated investments in facilities, data analytics and personnel to elevate supply-chain performance. [2]
- Management has publicly linked supply-chain improvements to better inventory metrics (out-of-stocks down year-over-year in key categories) and specifically said the company is working to use AI to improve demand forecasting to reduce volatility that can cause in-stock issues. [2]
- Recent leadership and organizational changes emphasize a store- and guest-experience focus that affects inventory priorities: new CEO Michael Fiddelke (Feb 2026) is reallocating investment toward stores and store staffing (adding payroll hours), while Target announced cuts of roughly 500 roles at distribution/regional offices to fund increased store investment; the company also made multiple senior supply-chain leadership moves including hiring Jeff England as Chief Supply Chain Officer. [10][8][9][2] ⚠
- Target is allocating material capital to turnaround efforts that include stores, merchandising and technology — management disclosed plans to invest about $5 billion in the coming year, including roughly $1 billion of incremental investment to improve stores, merchandise selection and digital capabilities. [12]
- Operational pain points and reputational risk tied to inventory and on-shelf availability are documented: media coverage cites customer complaints about messier shelves, out-of-stock items and longer checkout lines, and multiple rounds of layoffs and restructuring have been framed as efforts to reallocate resources to address these problems. [8][9][11]
- Target's technology leadership publicly promotes AI and generative AI as central to the company's roadmap (CIO commentary and corporate filings call out advancing technology and data capabilities), implying internal receptivity to AI-driven forecasting and data initiatives — but articles on enterprise AI note that data integration, explainability and retrieval of contextual data are common implementation challenges. [52][18][42] ⚠

Sources for Target

#	Source	Type	Published
2	Target moves to elevate supply chain operations, inventory reliability	News	2026-05-21
8	Target's New CEO Announces a Massive Overhaul That Finally Addresses Customers' Biggest Complaint	News	2026-02-10
9	Target cuts 500 jobs as retailer seeks to invest in its stores	News	2026-02-09
10	Target steps up investment in store staffing, cuts about 500 other roles	News	2026-02-09

11	Target cutting 800 jobs across Minneapolis and Brooklyn Park, raising concerns about broader economic slowdown | MPR News	News	2025-10-28
12	Target to Invest Billions to Upgrade Stores as Sales Slump Continues - WSJ	News	2025-11-19
18	Target CIO Brett Craig on Generative AI & Innovation Transforming Retail | Technovation 898	News	-
19	Brett Craig - Retired// Prior Chief Information Officer at ...	News	-
21	Leadership Team & Executive Officers | Target Corporation	News	-
22	Target Announces Executive Leadership Changes to Accelerate Growth, Confirms Q4 Financial Guidance	Leadership	-
24	Board of Directors & Management | Target Corporation	Leadership	-
25	Directors and Management	Leadership	-
27	Target Team Demand Planner Jobs, Employment	Jobs	-
30	Director of Data Sciences - Supply Chain Network ...	Jobs	-
36	Target Names Jeff England EVP, Chief Global Supply Chain and Logistics Officer	Leadership	-
37	Jeff England bio | Target Corporation	Leadership	-
38	Target Names New Chief Supply Chain and Logistics Officer	Leadership	-
39	Gretchen McCarthy - Fortune 50 Retail & Supply Chain ...	Leadership	-
42	Data management vendors race to connect AI with context | TechTarget	News	2026-07-28
46	o9 Solutions Taps Former Goldman Sachs and Target Exec as CTO - D CEO Magazine	Web	-
51	Tech Unicorn o9 Solutions Elevates Ashwin Rao To Chief Technology Officer - BW Disrupt	News	2026-07-03
52	TARGET CORP Form 10-K (filed 2026-03-11)	SEC filing	2026-03-11

Conclusion

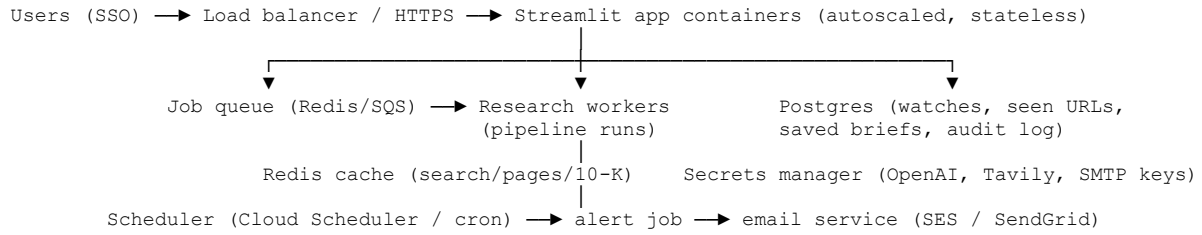
Building this project taught me four lessons:

- **Grounding matters more than the model.** Collecting evidence first and requiring a citation on every claim did more for accuracy than any prompt wording. The verifier then catches the remaining overstatements.
- **Small, focused agents are easier to improve.** When one section was weak (e.g. leader quotes), I could change one prompt and test it on its own, without affecting the others.
- **Data collection is the hard part.** Most of the debugging was in search, scraping and 10-K parsing, not in the LLM calls.
- **Measure, don't guess.** The one-pager only fit on one page once its printed height was measured and fed back to the writer (Experiment E2).

Appendix

Production deployment plan

Target architecture



The prototype runs everything in one process. In production I'd split it into three parts:

1. **Web tier.** The Streamlit container is stateless. It enqueues a research job and polls for the result, so a 2-minute run doesn't hold a web worker or fail when the browser reconnects.
2. **Worker tier.** Runs `run_pipeline` from the queue. It scales on queue depth, independently of web traffic.
3. **Scheduled alert job.** `cap-931-alerts` runs on a schedule (e.g. hourly) as a separate job.

Hosting: Docker images on Google Cloud Run, AWS ECS Fargate or Azure Container Apps. All three scale to zero when idle and can hold several instances under load. Streamlit Community Cloud is fine for the demo, but it lacks authentication and background workers.

Scalability

- **Most of the time goes to network and LLM calls, not CPU.**
  - Agents and page fetches already run in parallel threads.
  - Workers scale horizontally.
- **Shared Redis cache** instead of the local disk cache, so all workers reuse search, page and 10-K results. Popular accounts are researched once a day, not once per rep.
- **Rate limits.**
  - A per-provider token bucket covers OpenAI, Tavily and SEC (SEC allows 10 requests/s).
  - Exponential backoff is already in the OpenAI client (`max_retries=3`).
  - Use a paid search API (Tavily or similar) in production. DuckDuckGo's unofficial endpoint isn't meant for volume.

- **LLM cost control.**
  - The fast model does the high-token extraction work.
  - Per-user daily run quotas.
  - Usage is logged per run (already captured in `OnePager.usage`).
  - Prompt caching on the shared system prompt prefix.

Security

- **Secrets** (OpenAI, Tavily, SMTP) live in the cloud secrets manager and are injected as environment variables. They are never stored in the image or the repo (.env is gitignored).
- **Authentication.**
  - Put the app behind company SSO (OIDC via the load balancer or an auth proxy), so only reps can use it.
  - Alerts are tied to the signed-in user's email, not a free-text address.
- **Outbound fetching (SSRF).**
  - The app fetches user-supplied URLs, so in production block private IP ranges and metadata endpoints (169.254.169.254).
  - Cap response sizes, and allow only http/https.
  - Run workers in a network segment with no access to internal systems.
- **Prompt injection.**
  - Scraped text is wrapped in `<evidence>` tags and the system prompt marks it untrusted.
  - The verifier rejects claims the sources don't support.
  - No agent has tools that act on the outside world.
- **Data handling.**
  - Only public data is collected.
  - Uploaded decks are processed in memory and not stored.
  - Use an OpenAI organization with zero data retention for confidential decks.
- **Respect for sources.** Honor robots.txt, identify the client to SEC as required, and keep request rates modest.

Maintenance and operations

- **CI** (GitHub Actions):
  - `uv sync --locked`, `pytest` and a container build on every pull request.
  - Deploy to staging, then production on approval.
  - `uv.lock` pins dependencies, and Dependabot proposes updates.
- **Monitoring.**
  - Structured logs per run, with the run id, timings per stage, token usage per model and the number of sources.
  - Alerts on error rate, p95 latency and daily LLM spend.
- **Quality monitoring.**
  - Track the verifier's unsupported-claim rate over time. A rising rate is an early sign that a prompt or model change made things worse.
  - Re-run `experiments/run_experiments.py` before changing models or prompts.
- **Model lifecycle.**
  - Model names are configuration, not code.
  - When OpenAI retires a model, switch the environment variable, re-run the experiments and compare.
- **Scraper drift.**
  - Site markup changes over time. Extraction uses trafilatura, not site-specific selectors, and falls back to search snippets.
  - A weekly smoke test on a few known companies catches regressions.
- **Data retention.** Cached research expires after 24h. Saved briefs and alert history follow the company's retention policy.